Junior Deogracias MAMBO MOTSOU

Embedded Software & Hardware Engineer

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Profile

Embedded systems engineer (Polytech Dijon, 2026) working across firmware, electronics and digital hardware. At FORVIA I brought up firmware on new automotive silicon (LIN heater controller, MeLiBu LED nodes) and built the software of a nine-gateway lighting demonstrator. I work from datasheets, isolate faults to the hardware or software layer and validate on the bench; VHDL/FPGA design complements this.

Technical skills

Microcontrollers	STM32 (F446RE, G474RE), Infineon TLD4020 (Cortex-M23), Melexis MLX81116/MLX81143, RP2040 (PIO), PIC32MZ, PIC24, ATmega328P
Firmware	C (bare-metal, register-level CMSIS, HAL/SDK), FreeRTOS, Zephyr, interrupt and timer design, ADC acquisition, PWM, state machines
Buses & protocols	LIN, MeLiBu, UART, SPI, I²C, AXI4-Stream, Bluetooth serial, ISO 14443-A (MIFARE)
Digital design	VHDL (FSMs, pipelined datapaths, bidirectional IOBUF), Vivado (synthesis, implementation, IP integrator, ILA), Vitis HLS, XSim; Artix-7, Zynq-7000
Electronics	Schematic and PCB design (KiCad, Proteus), sensor interfacing (NTC, I²C/SPI sensors), evaluation-board bring-up
Debug & tools	J-Link, ST-LINK, PICkit on-board debugger, GDB, serial consoles; IAR EWARM, MPLAB X + Harmony v3/MCC, STM32CubeProgrammer, Pico SDK; Git, CMake, Ninja, West
Software	Python (PyQt5, OpenCV, threading), C++/Qt, Kotlin (Android, Jetpack Compose)

Professional experience

Embedded Systems Engineer — apprentice (contrat de professionnalisation)

Sep. 2025 – Sep. 2026

FORVIA – Faurecia Interior Systems Surface Activation department, France

Firmware bring-up: LIN-controlled heating-element controller (TCU)

- Developed C firmware on the Infineon TLD4020-3ET (Arm Cortex-M23 with integrated LIN): LIN ON/OFF command handling, heater PWM through the LDRV peripheral, NTC-based temperature regulation, fault protection and a state-machine architecture.
- Diagnosed early bring-up blockers: traced a HardFault in device initialisation to a silicon-revision mismatch on the evaluation board (register absent on older silicon), and a lost debug connection to boot-mode jumper and USB power conditions.
- The NTC divider is supplied from the switched heater output, so the ADC node is a 100 Hz square wave during PWM. Synchronised sampling to the PWM on-time (ISR at mid-pulse, 16-sample averaging), which brought measurement scatter within the regulation requirement.

Environment: TLD4020-3STD_EVAL board, IAR Embedded Workbench, Infineon MCU Configuration Wizard, on-board J-Link, printf over LIN (MobaXterm, 19 200 baud 8N1)

MeLiBu LED driver port: Melexis MLX81116 to MLX81143

- Ported an existing MeLiBu 1.x driver on RP2040 to the MLX81143: extended-mode sub-address encoding (14 LEDs per node), linear 15-bit PWM encoding confirmed against the datasheet and on the hardware, for a 22 × 7 panel (154 RGB LEDs).
- Fixed bus-timing faults: break generation with busy-wait while interrupts are masked, critical section narrowed to the break to stop UART RX byte loss, TX-FIFO drain to prevent back-to-back frame collisions; corrected the bus return pin.
- Isolated the panel's LED wiring scramble in the hardware-mapping layer, separate from the input convention; used a PIO-based receiver to confirm slave acknowledgements.

Environment: RP2040, Raspberry Pi Pico SDK, PIO state machines, C, MeLiBu 1.x

Interior ambient-lighting demonstrator (software owner)

- Extended the control software of an automotive interior demonstration buck from a few LED displays to nine STM32 UART gateways driving WS2812 LED strips across several lighting zones.
- Wrote a multithreaded Python/PyQt5 application: OpenCV decodes MP4 animations, maps logical LED IDs to frame pixels and distributes each frame to one worker thread per gateway at 921 600 baud, with a barrier to keep all nine gateways in step; custom serial frames (ASCII with delta compression).
- Built an Android tablet app (Kotlin, Jetpack Compose) that controls zones, colours and brightness presets over Bluetooth serial using a JSON command protocol with explicit error codes; added a keyboard fallback mode.
- Adapted the STM32 gateway firmware with minimal changes (ID-offset filtering when two gateways share a UART line).

Environment: Python 3, PyQt5, OpenCV, Kotlin/Jetpack Compose, STM32 (C), WS2812, UART, Bluetooth SPP, Git

Embedded Systems Intern Apr. – Aug. 2025
KMUTT — MDT Lab Bangkok, Thailand

- Image-processing work for a precision-agriculture project: fruit classification by transfer learning on VGG16 (Keras/TensorFlow), 98.4 % accuracy on a 4 599-image test set (Fruits-360, 131 classes), plus an OpenCV contour/colour detection pipeline.
- FPGA ramp-up: designed an 8-bit microcontroller in VHDL (12-state fetch–decode–execute controller, ALU with NZVC flags, memory-mapped ROM/RAM/I/O) following the LaMeres reference architecture; simulated in Vivado 2024.2 / XSim.

Environment: Python, TensorFlow/Keras, OpenCV, VHDL, Vivado 2024.2

Academic & personal projects

Polytech Dijon

Real-time thermometer: bare-metal PIC24, then FreeRTOS on PIC32MZ Polytech, solo then pair 2025 – 2026
github.com/teo1747/pic24-32-rt-thermometer

- V1 (solo): bare-metal PIC24FJ1024GB610 with RTCC timestamps and SPI EEPROM logging. V2 (pair): FreeRTOS pipeline of acquisition, analysis and display tasks on PIC32MZ2048EFH100, linked by queues and binary semaphores; timestamped messages; °C/°F toggle on a button.

Environment: Explorer 16/32, MPLAB X, XC16/XC32, Harmony v3/MCC, FreeRTOS 10.5, TC1047A sensor, on-board debugger

ZBT SRAM controller in VHDL Polytech SoC course, pair 2025 – 2026
github.com/teo1747/vhdl-zbt-sram-controller

- FSM controller for a Micron 512K × 36 ZBT SRAM: single and 4-word burst read/write on a 36-bit bidirectional bus (Xilinx IOBUF); self-checking testbench against the vendor memory model.

Environment: VHDL, Vivado, Zynq-7000 target, Micron simulation model

Real-time VGA video pipeline on ZedBoard Polytech, pair 2025 – 2026
github.com/teo1747/zedboard-vga-video-pipeline

- 640 × 480 chain built in IP integrator: Clocking Wizard, Video Timing Controller, AXI4-Stream video IPs, a VHDL inverter and Vitis HLS overlay IPs. Sync signals verified on hardware with ILA; the final demo showed no image because the HLS IPs had no AXI-Lite master to start them, since reworked as free-running (board retest pending).

Environment: Zynq-7000 (xc7z020), Vivado/Vitis HLS 2022.2, ILA, XDC constraints

3×3 image filter on FPGA Polytech, solo Nov. 2024 – Jan. 2025
github.com/teo1747/vhdl-2d-image-filter

- Streaming averaging and Gaussian (1-2-1/2-4-2/1-2-1) filters on 8-bit pixels: line buffers on the FIFO Generator IP, pipelined adder tree with explicit bit growth (9, 10, then 11 bits); simulated end-to-end on the Lena image, outputs compared with a software reference.

Environment: VHDL, Vivado 2024.1, Artix-7 (xc7a100t), XSim, MATLAB

Personal

Ongoing. I define the architecture, specifications and verification strategy; implementation is assisted by AI coding tools. Validation so far is in emulation (QEMU, EmbSIM), not on physical boards.

EmbLinkRTOS — preemptive RTOS for microcontrollers personal Oct. 2026 –
github.com/EmbLinkOs/EmbLinkRTOS

- Fixed-priority preemptive scheduler (bitmap ready lists, FIFO within a priority); Cortex-M port with PendSV/PSP context switch, lazy FPU save and BASEPRI critical sections; RISC-V (mret, CLINT) and AVR ports.
- Semaphores, mutexes with priority inheritance and priority ceiling, timeouts; conformance tests and differential testing against a Python reference model of the scheduler.
- CMake/Kconfig build with board definitions (NUCLEO-F446RE, Arduino Uno, ESP32-C3); GitHub Actions CI with ASan/UBSan/TSan, QEMU and EmbSIM runs.

RFID/PIN access-control terminal on FreeRTOS personal Oct. 2026
github.com/teo1747/stm32-freertos-access-control

- Register-level drivers (no vendor HAL) on STM32F446RE: MFRC522 RFID on SPI, SSD1306, DS3231 RTC and 24LC256 EEPROM on I²C, keypad, servo PWM, watchdog; six FreeRTOS tasks with queues, mutexes, event groups and stream buffers; salted, stretched SHA-256 PINs; CRC-checked log.
- Host unit tests with sanitizers and system scenarios in EmbSIM; not yet tested on a physical board.

Environment: NUCLEO-F446RE, FreeRTOS 11, CMSIS, arm-none-eabi-gcc, CMake, Ninja

Education & languages

Engineering degree (Diplôme d'ingénieur, Master's level) — Embedded Systems 2023 – Sep. 2026
Polytech Dijon Dijon, France
Graduated. Final-year apprenticeship at FORVIA. Coursework: electronics, PCB design, real-time systems, SoC/FPGA.

Two-year preparatory programme — mathematics, physics, electronics 2021 – 2023
IUC Cameroon

Languages English: native (anglophone), TOEIC C1 (2024) | French: fluent, professional use (B2)
Certification Cisco CCNA 1 (July 2024)